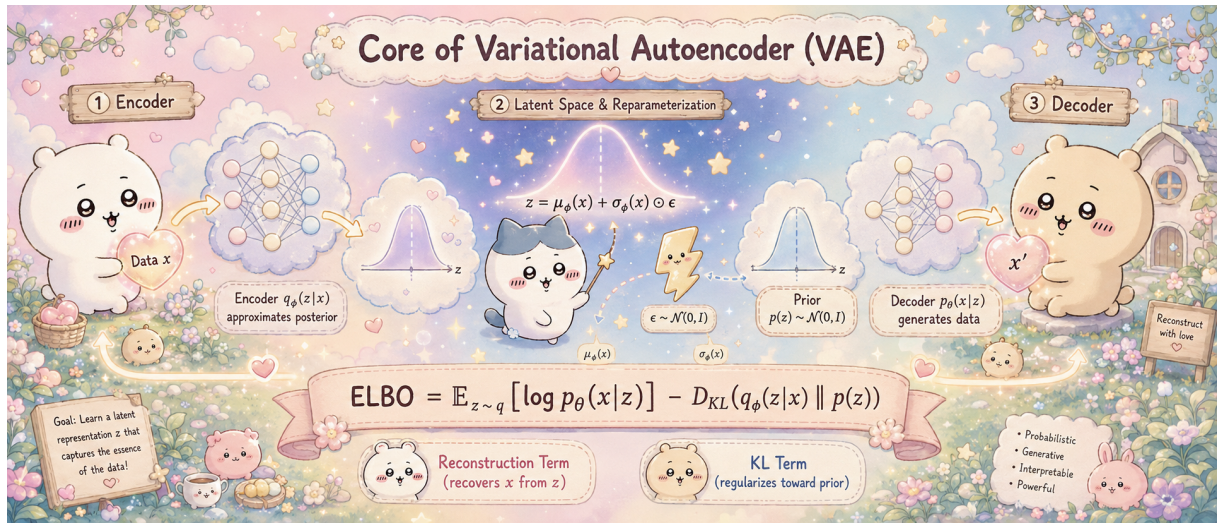


Chapter 2: Variational Autoencoder

2026 年 7 月 4 日

Deep Generative Models Notes

Model Structure	Why the Encoder	ELBO
Info-Theoretic View	Reparameterization	Gaussian VAE
Hierarchical VAE	HVAE ELBO	Pathologies



1 Model Structure

1.1 Encoder and Decoder

A VAE consists of two parameterized networks: encoder $q_\phi(z | \mathbf{x})$ and decoder $p_\theta(\mathbf{x} | z)$.

Definition 1: Variational Autoencoder

A **variational autoencoder** is a latent-variable generative model specified by

- a fixed **prior** $p(z)$ over a low-dimensional latent variable $z \in \mathbb{R}^d$;
- a **decoder** (generative model) $p_\theta(\mathbf{x} | z)$, a probabilistic map $\mathbb{R}^d \rightarrow \mathbb{R}^p$ parameterized by a neural network with weights θ ;
- an **encoder** (inference model) $q_\phi(z | \mathbf{x})$, a probabilistic map $\mathbb{R}^p \rightarrow \mathbb{R}^d$ parameterized by a neural network with weights ϕ , used to approximate the intractable posterior $p_\theta(z | \mathbf{x})$.

Information flow:

$$\mathbf{x} \xrightarrow{\text{encoder}} q_\phi(z | \mathbf{x}) \longrightarrow z \xrightarrow{\text{decoder}} p_\theta(\mathbf{x} | z) \longrightarrow \mathbf{x}'$$

Remark 2: From Deterministic to Probabilistic Autoencoders

A classical autoencoder learns a deterministic map $\mathbf{x} \mapsto z \mapsto \mathbf{x}'$ for reconstruction. A VAE instead makes the latent representation probabilistic and ties it to a prior $p(z)$, so new samples are generated by drawing $z \sim p(z)$ and decoding.

1.2 Variables

- $\mathbf{x}$: observed variable
- z : latent variable with prior $p_{\text{prior}}(z) = \mathcal{N}(z; \mathbf{0}, \mathbf{I})$
- Generation: sample $z \sim p_{\text{prior}}$, then $\mathbf{x} \sim p_{\theta}(\mathbf{x} | z)$

1.3 Marginal Likelihood

$$p_{\theta}(\mathbf{x}) = \int p_{\theta}(\mathbf{x} | z) p(z) dz = \mathbb{E}_{z \sim p(z)} [p_{\theta}(\mathbf{x} | z)].$$

This integral is intractable because $p_{\theta}(\mathbf{x} | z)$ is a nonlinear neural network, making the integral impossible to evaluate analytically.

2 Why We Need the Encoder

By Bayes' rule:

$$p_{\theta}(z | \mathbf{x}) = \frac{p_{\theta}(\mathbf{x} | z) p(z)}{p_{\theta}(\mathbf{x})}$$

Since $p_{\theta}(\mathbf{x})$ is intractable, so is the true posterior. We introduce a variational inference network:

$$q_{\phi}(z | \mathbf{x}) \approx p_{\theta}(z | \mathbf{x})$$

2.1 An Importance-Sampling Viewpoint

The encoder can be understood as a learned **proposal distribution** for importance sampling. Rewriting the marginal likelihood with an arbitrary $q_{\phi}(z | \mathbf{x}) > 0$,

$$p_{\theta}(\mathbf{x}) = \int q_{\phi}(z | \mathbf{x}) \frac{p_{\theta}(\mathbf{x} | z) p(z)}{q_{\phi}(z | \mathbf{x})} dz = \mathbb{E}_{z \sim q_{\phi}(z | \mathbf{x})} \left[\frac{p_{\theta}(\mathbf{x} | z) p(z)}{q_{\phi}(z | \mathbf{x})} \right].$$

Prior sampling is inefficient because most sampled z contribute little to explaining a fixed $\mathbf{x}$. The ideal proposal is the true posterior $p_{\theta}(z | \mathbf{x})$: then the importance weight $p_{\theta}(\mathbf{x} | z) p(z) / p_{\theta}(z | \mathbf{x}) = p_{\theta}(\mathbf{x})$ is a constant, so the estimator has zero variance and is maximally efficient. Since the true posterior is intractable, we learn $q_{\phi}(z | \mathbf{x})$ as a tractable approximation.

3 Training Objective: ELBO

3.1 Lower Bound

Definition 3: Evidence Lower Bound (ELBO)

For any datapoint $\mathbf{x}$ and any encoder $q_{\phi}(z | \mathbf{x})$ with the same support as the joint $p_{\theta}(\mathbf{x}, z)$, the **evidence lower bound** is

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) \triangleq \mathbb{E}_{z \sim q_{\phi}(z | \mathbf{x})} \left[\log \frac{p_{\theta}(\mathbf{x}, z)}{q_{\phi}(z | \mathbf{x})} \right] = \mathbb{E}_{z \sim q_{\phi}} [\log p_{\theta}(\mathbf{x} | z)] - D_{\text{KL}}(q_{\phi}(z | \mathbf{x}) \| p(z)).$$

The name reflects that the marginal log-likelihood $\log p_{\theta}(\mathbf{x})$ is often called the *evidence*.

Theorem 4: ELBO Lower Bound

$$\log p_\theta(\mathbf{x}) \geq \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) \triangleq \underbrace{\mathbb{E}_{z \sim q_\phi} [\log p_\theta(\mathbf{x} | z)]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p(z))}_{\text{latent KL}}$$

Proof. Since $\log p_\theta(\mathbf{x})$ does not depend on z , we may take an expectation over $z \sim q_\phi(z | \mathbf{x})$, then multiply and divide by $q_\phi(z | \mathbf{x})$ inside the logarithm:

$$\begin{aligned} \log p_\theta(\mathbf{x}) &= \mathbb{E}_{z \sim q_\phi} [\log p_\theta(\mathbf{x})] = \mathbb{E}_{z \sim q_\phi} \left[\log \frac{p_\theta(\mathbf{x}, z)}{p_\theta(z | \mathbf{x})} \right] \\ &= \mathbb{E}_{z \sim q_\phi} \left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z | \mathbf{x})} \right] + \underbrace{\mathbb{E}_{z \sim q_\phi} \left[\log \frac{q_\phi(z | \mathbf{x})}{p_\theta(z | \mathbf{x})} \right]}_{= D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p_\theta(z | \mathbf{x})) \geq 0} \\ &\geq \mathbb{E}_{z \sim q_\phi} \left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z | \mathbf{x})} \right] = \mathbb{E}_z [\log p_\theta(\mathbf{x} | z)] - D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p(z)), \end{aligned}$$

where the last line uses $p_\theta(\mathbf{x}, z) = p_\theta(\mathbf{x} | z) p(z)$. The same bound follows directly from Jensen's inequality applied to the concave log:

$$\log p_\theta(\mathbf{x}) = \log \mathbb{E}_{z \sim q_\phi} \left[\frac{p_\theta(\mathbf{x}, z)}{q_\phi(z | \mathbf{x})} \right] \geq \mathbb{E}_{z \sim q_\phi} \left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z | \mathbf{x})} \right].$$

□

Remark 5: Tightness of the Bound

The ELBO gap is

$$\log p_\theta(\mathbf{x}) - \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) = D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p_\theta(z | \mathbf{x})) \geq 0.$$

The bound is tight iff $q_\phi(z | \mathbf{x}) = p_\theta(z | \mathbf{x})$ almost everywhere.

3.2 Interpretation

- **Reconstruction term:** encourages recovery of $\mathbf{x}$ from z
- **KL term:** regularizes $q_\phi(z | \mathbf{x})$ toward the prior p_{prior}

4 Information-Theoretic View of ELBO

4.1 Two Joint Distributions

- **Generative joint:** $p_\theta(\mathbf{x}, z) = p(z) p_\theta(\mathbf{x} | z)$
- **Inference joint:** $q_\phi(\mathbf{x}, z) = p_{\text{data}}(\mathbf{x}) q_\phi(z | \mathbf{x})$

Proposition 6: Modeling Error Bound

$$D_{\text{KL}}(p_{\text{data}}(\mathbf{x}) \| p_\theta(\mathbf{x})) \leq D_{\text{KL}}(q_\phi(\mathbf{x}, z) \| p_\theta(\mathbf{x}, z))$$

Proof. Factor both joints by the chain rule, $q_\phi(\mathbf{x}, z) = p_{\text{data}}(\mathbf{x}) q_\phi(z | \mathbf{x})$ and $p_\theta(\mathbf{x}, z) =$

$p_\theta(\mathbf{x}) p_\theta(z | \mathbf{x})$, and split the logarithm of the ratio:

$$\begin{aligned} D_{\text{KL}}(q_\phi(\mathbf{x}, z) \| p_\theta(\mathbf{x}, z)) &= \mathbb{E}_{q_\phi(\mathbf{x}, z)} \left[\log \frac{p_{\text{data}}(\mathbf{x}) q_\phi(z | \mathbf{x})}{p_\theta(\mathbf{x}) p_\theta(z | \mathbf{x})} \right] \\ &= \mathbb{E}_{q_\phi(\mathbf{x}, z)} \left[\log \frac{p_{\text{data}}(\mathbf{x})}{p_\theta(\mathbf{x})} \right] + \mathbb{E}_{q_\phi(\mathbf{x}, z)} \left[\log \frac{q_\phi(z | \mathbf{x})}{p_\theta(z | \mathbf{x})} \right]. \end{aligned}$$

The first term depends only on $\mathbf{x}$, whose marginal under $q_\phi(\mathbf{x}, z)$ is $p_{\text{data}}(\mathbf{x})$, so it equals $D_{\text{KL}}(p_{\text{data}} \| p_\theta)$. In the second term, taking the inner expectation over $z \sim q_\phi(z | \mathbf{x})$ first turns each fiber into a KL divergence, leaving an outer expectation over $\mathbf{x} \sim p_{\text{data}}$:

$$\begin{aligned} D_{\text{KL}}(q_\phi(\mathbf{x}, z) \| p_\theta(\mathbf{x}, z)) &= D_{\text{KL}}(p_{\text{data}} \| p_\theta) + \mathbb{E}_{p_{\text{data}}} [D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p_\theta(z | \mathbf{x}))] \\ &\geq D_{\text{KL}}(p_{\text{data}} \| p_\theta), \end{aligned}$$

since the second term is an expectation of nonnegative KL divergences. $\square$

4.2 Error Decomposition

- $D_{\text{KL}}(p_{\text{data}} \| p_\theta)$: **modeling error**
- $\mathbb{E}_{p_{\text{data}}} [D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p_\theta(z | \mathbf{x}))]$: **inference error**

Remark 7: ELBO and Inference Error

$$\log p_\theta(\mathbf{x}) - \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) = D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p_\theta(z | \mathbf{x})) \geq 0$$

Maximizing ELBO therefore reduces inference error.

5 Reparameterization and Gradient Estimation

5.1 The Gradient Problem

Training maximizes $\mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x})]$ by stochastic gradient ascent. The gradient with respect to the decoder θ is straightforward, since the sampling distribution q_ϕ does not depend on θ :

$$\nabla_\theta \mathbb{E}_{z \sim q_\phi} [\log p_\theta(\mathbf{x} | z)] = \mathbb{E}_{z \sim q_\phi} [\nabla_\theta \log p_\theta(\mathbf{x} | z)].$$

The encoder gradient ∇_ϕ is harder because the sampling distribution itself depends on ϕ :

$$\nabla_\phi \mathbb{E}_{z \sim q_\phi(z | \mathbf{x})} [f(z)] \neq \mathbb{E}_{z \sim q_\phi(z | \mathbf{x})} [\nabla_\phi f(z)].$$

5.2 The Reparameterization Trick

Definition 8: Reparameterization

A distribution $q_\phi(z | \mathbf{x})$ is **reparameterizable** if there exist a fixed base distribution $p(\epsilon)$ (independent of ϕ) and a differentiable map $g_\phi(\epsilon, \mathbf{x})$ such that

$$z \sim q_\phi(z | \mathbf{x}) \iff z = g_\phi(\epsilon, \mathbf{x}), \quad \epsilon \sim p(\epsilon).$$

For the diagonal Gaussian encoder $q_\phi(z | \mathbf{x}) = \mathcal{N}(z; \mu_\phi(\mathbf{x}), \text{diag}(\sigma_\phi^2(\mathbf{x})))$ this is

$$z = \mu_\phi(\mathbf{x}) + \sigma_\phi(\mathbf{x}) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\odot$ denotes the elementwise product.

Proposition 9: Pathwise Gradient Estimator

If $q_\phi(z \mid \mathbf{x})$ is reparameterizable via $z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$, then for any differentiable f ,

$$\nabla_\phi \mathbb{E}_{z \sim q_\phi(z \mid \mathbf{x})} [f(z)] = \mathbb{E}_{\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})} [\nabla_\phi f(g_\phi(\boldsymbol{\epsilon}, \mathbf{x}))].$$

Proof. Because $p(\boldsymbol{\epsilon})$ does not depend on ϕ , the expectation operator and the gradient commute under the change of variables $z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$:

$$\nabla_\phi \mathbb{E}_{z \sim q_\phi(z \mid \mathbf{x})} [f(z)] = \nabla_\phi \mathbb{E}_{\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})} [f(g_\phi(\boldsymbol{\epsilon}, \mathbf{x}))] = \mathbb{E}_{\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})} [\nabla_\phi f(g_\phi(\boldsymbol{\epsilon}, \mathbf{x}))],$$

where the last step exchanges ∇_ϕ with the (now ϕ -free) expectation. The chain rule $\nabla_\phi f = (\partial_z f)(\partial_\phi g_\phi)$ then propagates the gradient through the sampled z . $\square$

5.3 Monte Carlo ELBO and Its Gradient

Combining reparameterization with a single Monte Carlo sample per datapoint gives the unbiased estimator actually used in practice. With $z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$ and $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$,

$$\widehat{\mathcal{L}_{\text{ELBO}}}(\theta, \phi; \mathbf{x}) = \log p_\theta(\mathbf{x} \mid g_\phi(\boldsymbol{\epsilon}, \mathbf{x})) - D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \parallel p(z)),$$

and its gradient (when the KL term is available in closed form, see Section 6) is

$$\nabla_{\theta, \phi} \widehat{\mathcal{L}_{\text{ELBO}}}(\theta, \phi; \mathbf{x}) = \nabla_{\theta, \phi} [\log p_\theta(\mathbf{x} \mid g_\phi(\boldsymbol{\epsilon}, \mathbf{x})) - D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \parallel p(z))],$$

which is an unbiased estimator of $\nabla_{\theta, \phi} \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x})$ and is computed by ordinary backpropagation.

Remark 10: Why Not the Score-Function Estimator

The general-purpose score-function (REINFORCE) estimator

$$\nabla_\phi \mathbb{E}_{z \sim q_\phi} [f(z)] = \mathbb{E}_{z \sim q_\phi} [f(z) \nabla_\phi \log q_\phi(z \mid \mathbf{x})]$$

is unbiased and does not require reparameterization, but often has much higher variance. For continuous latents, pathwise gradients are typically more stable in practice.

6 Gaussian VAE

6.1 Parameterization

Encoder (diagonal Gaussian):

$$q_\phi(z \mid \mathbf{x}) \triangleq \mathcal{N}(z; \boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x})))$$

where $\boldsymbol{\mu}_\phi : \mathbb{R}^p \rightarrow \mathbb{R}^d$, $\boldsymbol{\sigma}_\phi : \mathbb{R}^p \rightarrow \mathbb{R}_+^d$.

Decoder (isotropic Gaussian):

$$p_\theta(\mathbf{x} \mid z) \triangleq \mathcal{N}(\mathbf{x}; \mathbf{M}_\theta(z), \sigma^2 \mathbf{I})$$

where $\mathbf{M}_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^p$ and $\sigma > 0$ is a fixed small constant.

6.2 Equivalent Optimization

Expanding the reconstruction term:

$$\mathbb{E}_z[\log p_\theta(\mathbf{x} | z)] = -\frac{1}{2\sigma^2} \mathbb{E}_z[\|\mathbf{x} - \mathbf{M}_\theta(z)\|^2] + \text{const}$$

So maximizing ELBO is equivalent to:

$$\min_{\theta, \phi} \mathbb{E}_{p_{\text{data}}} \left[\mathbb{E}_{q_\phi} \left[\frac{1}{2\sigma^2} \|\mathbf{x} - \mathbf{M}_\theta(z)\|^2 \right] + D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p_{\text{prior}}(z)) \right]$$

Here σ controls the reconstruction–regularization trade-off. The reconstruction term is estimated via reparameterized sampling (Section 5), while the KL term is closed form.

Proposition 11: Closed-Form Gaussian KL

When both q_ϕ and p_{prior} are Gaussian with diagonal covariance,

$$D_{\text{KL}}(\mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) = \frac{1}{2} \sum_{j=1}^d (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1).$$

Proof. Because both distributions factorize over the d coordinates, the KL divergence is the sum of the per-coordinate divergences; it suffices to prove the one-dimensional case for $q = \mathcal{N}(\mu, \sigma^2)$ and $p = \mathcal{N}(0, 1)$. Writing out the densities,

$$\log \frac{q(z)}{p(z)} = \log \frac{1}{\sigma} - \frac{(z - \mu)^2}{2\sigma^2} + \frac{z^2}{2} = -\frac{1}{2} \log \sigma^2 - \frac{(z - \mu)^2}{2\sigma^2} + \frac{z^2}{2}.$$

Taking the expectation over $z \sim q$ and using $\mathbb{E}_q[(z - \mu)^2] = \sigma^2$ together with $\mathbb{E}_q[z^2] = \mu^2 + \sigma^2$,

$$D_{\text{KL}}(q \| p) = \mathbb{E}_q \left[\log \frac{q(z)}{p(z)} \right] = -\frac{1}{2} \log \sigma^2 - \frac{1}{2} + \frac{\mu^2 + \sigma^2}{2} = \frac{1}{2} (\mu^2 + \sigma^2 - \log \sigma^2 - 1).$$

Summing over the d independent coordinates yields the stated formula. $\square$

Remark 12: A Fully Differentiable Objective

With reparameterized reconstruction and closed-form KL (Proposition 11), the per-sample objective is fully differentiable in (θ, ϕ) and optimized by standard backpropagation.

7 Hierarchical VAE (HVAE)

7.1 Motivation

Hierarchical VAEs introduce a hierarchy of latent variables, allowing the model to capture features at multiple abstraction levels.

7.2 Hierarchical Encoder-Decoder Chain

$$\mathbf{x} \xrightarrow[q_\phi(z_1 | \mathbf{x})]{p_\theta(\mathbf{x} | z_1)} z_1 \xrightarrow[q_\phi(z_2 | z_1)]{p_\theta(z_1 | z_2)} \dots \xrightarrow[q_\phi(z_L | z_{L-1})]{p_\theta(z_{L-1} | z_L)} z_L$$

7.3 Modeling

The generative joint distribution is:

$$p_\theta(\mathbf{x}, z_{1:L}) = p_\theta(\mathbf{x} | z_1) \prod_{i=2}^L p_\theta(z_{i-1} | z_i) p(z_L)$$

The marginal data distribution becomes:

$$p_{\text{HVAE}}(\mathbf{x}) = \int p_{\theta}(\mathbf{x}, z_{1:L}) dz_{1:L}$$

7.4 Generation and Encoding

Generation samples top latent $z_L \sim p(z_L)$, then decodes downward layer by layer to z_1 , and finally generates $\mathbf{x}$ from $p_{\theta}(\mathbf{x} | z_1)$.

Encoding follows a Markov factorization:

$$q_{\phi}(z_{1:L} | \mathbf{x}) = q_{\phi}(z_1 | \mathbf{x}) \prod_{i=2}^L q_{\phi}(z_i | z_{i-1})$$

This bottom-up Markov chain mirrors the top-down decoder and keeps inference/sampling tractable through simple conditional factors.

8 HVAE ELBO

8.1 Derivation

$$\begin{aligned} \log p_{\text{HVAE}}(\mathbf{x}) &= \log \int p_{\theta}(\mathbf{x}, z_{1:L}) dz_{1:L} \\ &= \log \int q_{\phi}(z_{1:L} | \mathbf{x}) \frac{p_{\theta}(\mathbf{x}, z_{1:L})}{q_{\phi}(z_{1:L} | \mathbf{x})} dz_{1:L} \\ &= \log \mathbb{E}_{q_{\phi}(z_{1:L} | \mathbf{x})} \left[\frac{p_{\theta}(\mathbf{x}, z_{1:L})}{q_{\phi}(z_{1:L} | \mathbf{x})} \right] \\ &\geq \mathbb{E}_{q_{\phi}(z_{1:L} | \mathbf{x})} \left[\log \frac{p_{\theta}(\mathbf{x}, z_{1:L})}{q_{\phi}(z_{1:L} | \mathbf{x})} \right] \triangleq \mathcal{L}_{\text{ELBO}}^{\text{HVAE}}(\theta, \phi; \mathbf{x}) \end{aligned}$$

8.2 Three-Term Decomposition

Substituting the generative and inference factorizations into $\log \frac{p_{\theta}(\mathbf{x}, z_{1:L})}{q_{\phi}(z_{1:L} | \mathbf{x})}$ and using $z_0 \equiv \mathbf{x}$ yields a reconstruction term, adjacent-layer KL matching terms, and a top-level KL to the prior:

$$\begin{aligned} \mathcal{L}_{\text{ELBO}}^{\text{HVAE}}(\theta, \phi; \mathbf{x}) &= \underbrace{\mathbb{E}_{q_{\phi}(z_1 | \mathbf{x})} [\log p_{\theta}(\mathbf{x} | z_1)]}_{\text{reconstruction term}} \\ &\quad - \underbrace{\sum_{i=1}^{L-1} \mathbb{E}_{q_{\phi}(z_{1:L} | \mathbf{x})} [D_{\text{KL}}(q_{\phi}(z_i | z_{i-1}) \| p_{\theta}(z_i | z_{i+1}))]}_{\text{adjacent-layer matching terms}} \\ &\quad - \underbrace{\mathbb{E}_{q_{\phi}(z_{L-1} | \mathbf{x})} [D_{\text{KL}}(q_{\phi}(z_L | z_{L-1}) \| p(z_L))]}_{\text{top-level KL regularizer}} \end{aligned}$$

where $z_0 \equiv \mathbf{x}$ for notational convenience.

9 Pathologies of Deep Networks in Flat VAE

9.1 Reason 1: Deeper Networks Do Not Expand the Variational Family

In the standard variational family

$$q_{\phi}(z | \mathbf{x}) = \mathcal{N}(z; \mu_{\phi}(\mathbf{x}), \text{diag}(\sigma_{\phi}^2(\mathbf{x}))),$$

deeper encoder networks may improve parameter fitting quality, but they do not change the family itself. If the true posterior $p_\theta(z | \mathbf{x})$ is multi-modal, a diagonal Gaussian approximation remains fundamentally limited.

9.2 Reason 2: Posterior Collapse

The expected ELBO over data can be decomposed as:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\mathcal{L}_{\text{ELBO}}(\mathbf{x})] = \mathbb{E}_{p_{\text{data}}(\mathbf{x}) q_\phi(z|\mathbf{x})} [\log p_\theta(\mathbf{x} | z)] - I_\phi(\mathbf{x}; z) - D_{\text{KL}}(\bar{q}_\phi(z) \| p(z)),$$

where the aggregated posterior is

$$\bar{q}_\phi(z) \triangleq \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [q_\phi(z | \mathbf{x})],$$

and the mutual information is

$$I_\phi(\mathbf{x}; z) \triangleq \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| \bar{q}_\phi(z))].$$

This decomposition follows by averaging the per-datapoint latent KL and inserting the aggregated posterior $\bar{q}_\phi(z)$ inside the logarithm:

$$\begin{aligned} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p(z))] &= \mathbb{E}_{p_{\text{data}}(\mathbf{x}) q_\phi(z|\mathbf{x})} \left[\log \frac{q_\phi(z | \mathbf{x})}{\bar{q}_\phi(z)} + \log \frac{\bar{q}_\phi(z)}{p(z)} \right] \\ &= \underbrace{\mathbb{E}_{p_{\text{data}}(\mathbf{x})} [D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| \bar{q}_\phi(z))]}_{= I_\phi(\mathbf{x}; z)} + \underbrace{\mathbb{E}_{\bar{q}_\phi(z)} \left[\log \frac{\bar{q}_\phi(z)}{p(z)} \right]}_{= D_{\text{KL}}(\bar{q}_\phi(z) \| p(z))}, \end{aligned}$$

where the second term uses that the marginal of z under $p_{\text{data}}(\mathbf{x}) q_\phi(z | \mathbf{x})$ is exactly $\bar{q}_\phi(z)$. Substituting back into $\mathbb{E}_{p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}] = \mathbb{E}_{p_{\text{data}} q_\phi} [\log p_\theta(\mathbf{x} | z)] - \mathbb{E}_{p_{\text{data}}} [D_{\text{KL}}(q_\phi(z | \mathbf{x}) \| p(z))]$ gives the three-term form above.

If the decoder is too expressive, the model class may contain solutions with $p_\theta(\mathbf{x} | z) \approx r(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$ that almost ignore z . Then $q_\phi(z | \mathbf{x})$ is pushed toward $p(z)$, implying $I_\phi(\mathbf{x}; z) \rightarrow 0$ and KL penalties close to zero.

Remark 13: Consequences of Posterior Collapse

- z becomes nearly independent of $\mathbf{x}$, carrying little useful information.
- Conditioning on z or moving in latent space has little effect on generated samples.

9.3 What Hierarchy Changes?

One structural response to posterior collapse is to adopt hierarchical latents (HVAE; Section 7). Instead of routing all information through a single code, the model distributes representation across multiple levels, with adjacent-layer matching terms that constrain inference and generation at every interface.

The HVAE ELBO (Section 8) can be written in log-difference form as

$$\begin{aligned} \mathcal{L}_{\text{ELBO}}^{\text{HVAE}}(\theta, \phi; \mathbf{x}) &= \mathbb{E}_{q_\phi(z_1|\mathbf{x})} [\log p_\theta(\mathbf{x} | z_1)] \\ &\quad - \mathbb{E}_{q_\phi(z_{1:2}|\mathbf{x})} [\log q_\phi(z_1 | \mathbf{x}) - \log p_\theta(z_1 | z_2)] \\ &\quad - \sum_{i=2}^{L-1} \mathbb{E}_{q_\phi(z_{1:i+1}|\mathbf{x})} [\log q_\phi(z_i | z_{i-1}) - \log p_\theta(z_i | z_{i+1})] \\ &\quad - \mathbb{E}_{q_\phi(z_{L-1}|\mathbf{x})} [D_{\text{KL}}(q_\phi(z_L | z_{L-1}) \| p(z_L))]. \end{aligned}$$

Under $z_0 \equiv \mathbf{x}$, each log-difference term is exactly the corresponding adjacent-layer KL matching term in Section 8; the top-level term is already a KL.

Remark 14: Layer-Wise Abstraction in HVAE

Each latent level interacts only with its neighbors: the encoder passes information upward through $q_\phi(z_i \mid z_{i-1})$, and the decoder passes information downward through $p_\theta(z_{i-1} \mid z_i)$. These properties stem from the hierarchical latent graph, not from simply deepening networks in a flat VAE. Because matching penalties act at every layer, a powerful decoder cannot ignore the entire latent hierarchy without incurring multiple interface costs; HVAE is therefore typically less collapse-prone than a flat VAE with a deeper encoder alone.